

TNL6323 Natural Language Processing
Trimester March/April 2026
Project (40%)

Description

This is a group project which carries 40% of your coursework for TNL6323 Natural Language Processing. You are to work in a group of at most **4** students from the same lab section. All group members must contribute to the development and documentation of the project. You may develop the application using any programming language and libraries.

Malaysia-Focused Sentiment Analysis System (Application-Based)

Sentiment Analysis (SA) is the task of detecting positive, neutral or negative sentiment from text. In this project, you will develop an SA system, which involves several steps, from data collection and pre-processing to model training and evaluation. This guideline outlines possible steps to complete the project (not limited to these). Choose **ONE** Malaysia-specific domain below for your sentiment analysis (SA) system. Each domain is contextualized to reflect Malaysian culture, platforms, habits, and local trends. There is **NO REPEATED** title that can be selected in same lab section. The title selection is based on first come first serve basics.

- a. Movie/ Entertainment Reviews (Malaysia Context): Sentiment analysis of Malaysian audiences toward local movies, cinema experiences (GSC, TGV), Astro shows, influencer content, and reactions to new releases (e.g., Malay dramas and local films).
- b. Product/ E-commerce Reviews (Malaysia Context): Customer reviews from Shopee Malaysia, Lazada Malaysia and feedback on popular products categories including skincare, home appliances and etc.
- c. Social Media & Public Opinion (Malaysia Trends): Posts from Twitter Malaysia, Instagram and TikTok related to local trends such as Hari Raya sales, festivals and national events.
- d. Restaurant & Food Reviews (Malaysia Context): Sentiment analysis of Malaysian dining experiences, including mamak stalls, kopitiam and restaurants based on reviews from Google Maps, GrabFood and Foodpanda.
- e. Healthcare Feedback (Malaysia Context): Public sentiment toward Malaysian healthcare services, including hospitals (e.g., Pantai, Putra, government hospitals), clinics, dental services and traditional medicine providers.
- f. News Articles (Malaysia Context): Sentiment analysis of local news related to finance, politics and society from sources such as Malaysiakini, The Star and Berita Harian.
- g. Hotel & Travel Reviews (Malaysia Tourism): Reviews of hotels, airlines, tourist attractions (e.g., Langkawi, Melaka, Penang) from platforms like Booking.com and Agoda, focusing on domestic travel experiences.
- h. Business & Marketing (Malaysia Brands): Customer feedback and public reactions to campaigns by local brands (e.g., Grab campaigns, Touch 'n Go eWallet promotions and SME branding efforts).
- i. Technology (Malaysia Tech Scene): Public opinions on emerging technologies, AI tools, smartphones and telecommunications services (e.g., CelcomDigi, Maxis, U Mobile) including reactions to local tech developments.
- j. App Reviews (Malaysia Usage): User reviews of widely used mobile applications in Malaysia such as eWallet, Grab, MySejahtera, banking apps and government service apps.

Steps	Description
1. Data Collection	Choose a domain from webpages, social media, and etc. Collect at least 360 sentences (120 positive, 120 neutral, 120 negative). You may use APIs and web scraping techniques.
2. Data Pre-processing	Pre-process the data by performing text cleaning, tokenization, stemming, translation, and etc (whichever applicable).
3. Knowledge Representation	Represent knowledge in an appropriate format such as csv, hdf5, and etc.
4. Model Training	Choose and train a model on data using traditional machine learning models or deep learning models.
5. Deployment	Deploy the sentiment analysis system on a platform or interface that users can access, such as a web application or an API.

Advanced Features (including but not limited to)	Description
Emoji/ Emoticon Sentiment Integration	Detect and interpret emojis or emoticons in text to capture the expressed sentiment.
Transformer Model	Incorporate a modern transformer-based deep learning model to enhance prediction accuracy.
Aspect-based Sentiment Analysis	Automatically identify and categorize different aspects of text (e.g., food, service, ambiance for restaurant reviews or product quality, delivery, packaging for e-commerce reviews)

Submission and Presentation

- Please specify your group members in the shared Excel sheet on eBwise.
- The **project leader** is required to zip all the files and upload them to **eBwise**. Name the zip file as the lab section and group number, e.g.: **Lab1A-G1.zip**. Below are the files to be submitted:
 - Data (cleaned), and source codes of the Sentiment Analysis system. (with two folders: /Data and /Source Codes)
 - Report in word and pdf
 - User manual (explanation of how to install and use your source code)
 - Group Project Contribution Form
 - A recorded video of system demo with explanation (at most 5 minutes). All members must present in the video.
- The submission is to be completed at the latest by **03 July 2026 (Friday), 5pm**. Penalty will be imposed for late submission.

**TNL6323 Natural Language Processing
Project Marking Rubrics**

No.	Marking Scheme	Marks
1.	System (25 marks) • Data Collection – 4% • Data Pre-processing – 4% • Knowledge Representation – 5% • Model Training – 5% • Deployment – 3% • Advanced Features – 3% • System Demo – 1%	
2.	Final Report (15 marks) • Introduction - 1% • Literature Review (at least 12 related works, published within the year 2021-2026) - 1% • Data Collection – 2% • Data Pre-processing – 2% • Knowledge Representation - 2% • Model Training – 2% • Deployment (with screenshots) – 2% • Advanced Features – 2% • Formatting - 1% <i>[Important: AI & Turnitin similarity score > 15% minus 5 marks]</i>	
TOTAL		/ 40

Useful Resources (Not limited to these)

1. <https://huggingface.co/blog/sentiment-analysis-python>
2. <https://github.com/RichardRivaldo/Sentiment-Analysis>
3. <https://medium.com/@swayampatil7918/getting-started-with-sentiment-analysis-a-step-by-step-guide-1a16085688a7>